Supplementary Material

Plane Conditioning, Train–Test Leakage and External Validity in Joint Brain Tumor Classification and Segmentation

S1. CLAIM Checklist

Checklist for Artificial Intelligence in Medical Imaging (CLAIM) [1]. Section names refer to the main manuscript.

#	Item	Where / how addressed
Title and abstract		
1	Identification as AI study, category of technology	Title; Abstract (deep learning, convolutional encoder–decoder).
2	Structured summary	Abstract (design, data, methods, results, conclusion).
Introduction		
3	Scientific and clinical background, intended use	Introduction, paragraphs 1–3. Research benchmark study; no clinical deployment is intended.
4	Objectives and hypotheses	Introduction (contributions); pre-specified hypotheses, margins and endpoints in the analysis plan (Statistical Analysis).
Methods: study design		
5	Prospective or retrospective	Retrospective analysis of public datasets (Materials).
6	Study goal	Model development with controlled comparison, leakage audit and external validation (Introduction).
Methods: data		
7	Data sources	BRISC2025, PMRAM, AJBDS-2023, with references and DOIs (Materials; Data and Code Availability).
8	Eligibility criteria	All images of the public releases; PMRAM exact duplicates removed; AJBDS-2023 slices with a paired mask (Materials). Patient-level clinical data are not available.
9	Pre-processing	Resizing to 224×224, ImageNet normalisation, binarisation of masks (Training).
10	Selection of data subsets	Leakage-free development subset (Leakage-Free Retraining); de-duplicated and strict test subsets (Materials).
11	Definitions of data elements	Four classes and binary tumor masks as defined by the dataset providers (Materials).
12	De-identification	Performed by the dataset providers; no identifiable data used (Materials).
13	Missing data	Healthy BRISC2025 slices have no mask file and are assigned an empty mask (Training); AJBDS-2023 images without a mask are excluded (Materials).
Methods: ground truth		
14	Reference standard	Expert tumor masks and class labels released with each dataset (Materials).
15	Rationale for reference standard	Only available reference; differences in lesion definition between cohorts are discussed (Discussion).

#	Item	Where / how addressed
16	Source and qualifications of annotators	As reported by the dataset providers [2, 3]; not re-annotated.
17	Annotation tools	As reported by the dataset providers.
18	Inter-/intra-rater variability	Not available for the released masks; acknowledged in Limitations.
Methods: data partitions		
19	Sample size	Determined by the public releases; no power calculation. Precision is reported through confidence intervals and the equivalence analysis.
20	Assignment to partitions	Official BRISC2025 train/test split; seeded 80/20 train/validation split stratified by class × plane (Materials).
21	Level of disjointness	Image level; patient level impossible (no identifiers). Near-duplicate audit and leakage-free retraining quantify the consequence (Materials; Results). AJBDS-2023 fine-tuning is patient-disjoint.
Methods: model		
22	Model description	Network (Methods; Fig. 1).
23	Software	Python 3.11, PyTorch 2.11 (CUDA 12.8), torchvision 0.26, NumPy 2.4, SciPy 1.17, scikit-learn 1.8, pandas 3.0, Pillow 12.2.
24	Initialisation	ImageNet-pretrained ResNet-34; plane embedding $\mathcal{N}(0, 0.02^2)$; fine-tuning initialised from the BRISC2025 models (Methods).
Methods: training		
25	Training details	Training (loss, optimiser, schedule, augmentation, batch size, epochs); number of models: Table S1.
26	Final model selection	Highest validation score $0.5\,F1_{\text{macro}} + 0.5\,\text{Dice}$; no test-set use (Training).
27	Ensembling	Averaging of softmax and pixel probabilities over five seeds (Training).
Methods: evaluation		
28	Metrics	Metrics (classification, segmentation, calibration).
29	Statistical measures	Statistical Analysis (bootstrap CIs, Wilcoxon, McNemar, Holm, TOST equivalence, patient-cluster bootstrap).
30	Robustness / sensitivity analysis	De-duplicated and strict test subsets, leakage-free retraining, zero- vs mean-plane inference, threshold sweeps (Results).
31	Explainability and its validation	LayerCAM with quantitative localisation (energy in mask, pointing game) (Explainability).
32	External testing	PMRAM and AJBDS-2023, zero-shot and fine-tuned (External Protocols; Results).
Results		
33	Flow of cases	Fig. 2; exclusion counts in Materials and Table S1.
34	Demographic/clinical characteristics	Class and plane distribution per partition (Materials; Fig. 2). Patient demographics are not available for BRISC2025 and PMRAM.
35	Performance on all partitions	Test-set performance for every model (Table S2); validation NLL and temperatures (Table S3).
36	Precision of estimates	95% CIs throughout (Results; Tables S2–S4).
37	Failure analysis	Lower-decile cases (Fig. 5), confusion matrix (Fig. 6C), subgroup analyses, external failure modes (Results).
Discussion		
38	Limitations	Limitations.

#	Item	Where / how addressed
39	Implications for practice	Discussion (no clinical use without target-domain adaptation and prospective validation).
Other information		
40	Registration	Not registered. The analysis plan for the revision experiments was fixed in writing before those runs (repository: `reports/revision2_analysis_plan.md`).
41	Protocol access	Released with the code (Data and Code Availability).
42	Funding	See Acknowledgment.

S2. Supplementary Tables

Model names: B2 = joint model without plane embedding; B3 = with plane embedding; s = training seed; LF = leakage-free retraining split; ens. = 5-seed ensemble.

Table S1: Training runs. Leakage-free split: 394 development images matched to a test image and 38 redundant internal copies removed (425 in total). Train/validation sizes are approximate for the leakage-free split (stratified rounding).

Setting	Models	Seeds	Dev. images	Train / val	Purpose
Official split	B2, B3	42, 43, 44, 45, 46	5000	4000 / 1000	Main comparison
Leakage-free split	B2, B3	42, 43, 44, 45, 46	4575	3660 / 915	Leakage-free comparison
Official split	B0, B1, A2, A3, A5, B3 replicate	42	5000	4000 / 1000	Single-run ablations
AJBDS-2023 fine-tuning	B2, B3 (from seed 42)	—	13–14 patients	5 folds × 4 sizes	Target-domain adaptation

Table S2: Internal test results of every model (BRISC2025 official test set, $N = 1{,}000$). Percentages; HD95 in pixels (median / mean over tumor slices with a non-empty prediction).

Model	Acc.	F1	AUC	Tumor Dice	All Dice	IoU	Sens.	Prec.	HD95	Missed	Healthy FP
B2 s42	99.40	99.47	99.92	85.88	87.86	78.36	87.38	87.67	2.24 / 5.91	5	0
B2 s43	99.40	99.47	99.89	86.35	88.26	78.98	87.69	87.91	2.24 / 6.27	5	0
B2 s44	99.70	99.74	99.90	85.91	87.89	78.49	87.66	87.15	2.24 / 6.29	5	0
B2 s45	99.10	99.22	99.79	85.93	87.90	78.56	87.56	87.35	2.24 / 6.04	6	0
B2 s46	99.40	99.47	99.93	86.25	88.17	78.78	87.63	87.87	2.24 / 5.98	6	0
B2 ens.	99.60	99.65	99.94	86.51	88.40	79.22	87.83	88.22	2.24 / 5.73	7	0
B3 s42	99.30	99.39	99.91	86.27	88.20	78.80	88.51	86.99	2.24 / 6.00	4	0
B3 s43	99.40	99.48	99.94	85.96	87.93	78.52	87.53	87.61	2.24 / 6.02	7	0
B3 s44	99.20	99.26	99.94	85.67	87.67	78.20	87.36	87.30	2.24 / 6.23	8	0
B3 s45	99.50	99.57	99.98	85.66	87.66	78.26	87.90	86.74	2.24 / 6.21	8	0
B3 s46	99.20	99.26	99.97	85.77	87.76	78.24	87.67	87.19	2.24 / 6.23	6	0
B3 ens.	99.60	99.65	99.95	86.31	88.22	79.01	88.08	87.68	2.24 / 5.85	7	0
B2 LF s42	97.10	96.86	99.87	83.50	85.81	75.88	85.88	85.94	2.83 / 6.99	17	0
B2 LF s43	97.20	97.13	99.67	84.48	86.65	76.92	85.99	87.36	2.44 / 6.61	16	0
B2 LF s44	97.70	97.51	99.73	84.85	86.97	77.40	86.20	87.80	2.38 / 6.11	12	0
B2 LF s45	97.70	97.62	99.36	84.94	87.05	77.42	86.56	87.64	2.24 / 6.55	15	0
B2 LF s46	98.90	98.89	99.68	85.31	87.37	77.74	86.10	88.15	2.24 / 6.52	9	0
B2 LF ens.	98.10	97.95	99.89	85.22	87.29	77.81	86.43	88.17	2.24 / 6.03	15	0
B3 LF s42	98.20	98.21	99.91	84.43	86.61	76.88	86.34	86.91	2.38 / 6.53	17	0
B3 LF s43	98.30	98.18	99.81	84.41	86.60	76.70	85.82	87.12	2.83 / 6.84	12	0
B3 LF s44	97.30	97.13	99.28	84.22	86.43	76.65	85.75	87.48	2.59 / 6.64	18	0
B3 LF s45	97.80	97.66	99.57	84.54	86.70	76.95	86.46	86.67	2.83 / 6.44	13	0
B3 LF s46	97.00	96.74	99.70	84.16	86.38	76.68	85.89	86.92	2.68 / 6.54	17	0
B3 LF ens.	98.30	98.15	99.85	85.12	87.20	77.67	86.58	87.92	2.24 / 5.90	17	0

Table S3: Temperature scaling. T fitted on each model's own validation split; test-set ECE (15 bins), NLL and Brier before and after scaling.

Model	n_{val}	T	Val NLL	Val NLL (TS)	ECE	ECE (TS)	NLL	NLL (TS)	Brier	Brier (TS)
B2 s42	1000	0.553	0.0994	0.0368	0.0756	0.0013	0.1001	0.0361	0.0186	0.0119
B3 s42	1000	0.556	0.0989	0.0365	0.0740	0.0021	0.1030	0.0410	0.0200	0.0133
B2 LF s42	915	0.512	0.0870	0.0227	0.0634	0.0244	0.1533	0.1437	0.0523	0.0538
B3 LF s42	915	0.547	0.0941	0.0319	0.0663	0.0119	0.1351	0.0998	0.0373	0.0336
B2 s43	1000	0.564	0.1028	0.0427	0.0743	0.0010	0.0998	0.0392	0.0187	0.0121
B3 s43	1000	0.605	0.1082	0.0556	0.0727	0.0048	0.1004	0.0397	0.0193	0.0123
B2 LF s43	915	0.584	0.1009	0.0450	0.0664	0.0193	0.1608	0.1406	0.0520	0.0510
B3 LF s43	915	0.560	0.0983	0.0375	0.0699	0.0114	0.1342	0.0947	0.0373	0.0333
B2 s44	1000	0.534	0.0946	0.0296	0.0781	0.0016	0.0928	0.0226	0.0141	0.0062
B3 s44	1000	0.551	0.0950	0.0337	0.0716	0.0038	0.1053	0.0504	0.0219	0.0159
B2 LF s44	915	0.525	0.0901	0.0253	0.0680	0.0182	0.1431	0.1181	0.0438	0.0423
B3 LF s44	915	0.523	0.0900	0.0252	0.0677	0.0217	0.1583	0.1449	0.0504	0.0492
B2 s45	1000	0.551	0.0964	0.0344	0.0736	0.0048	0.1073	0.0524	0.0232	0.0179
B3 s45	1000	0.533	0.0932	0.0283	0.0745	0.0011	0.0972	0.0327	0.0170	0.0099
B2 LF s45	915	0.571	0.0986	0.0406	0.0649	0.0164	0.1513	0.1252	0.0465	0.0432
B3 LF s45	915	0.545	0.0938	0.0312	0.0605	0.0180	0.1529	0.1337	0.0464	0.0433
B2 s46	1000	0.551	0.0942	0.0333	0.0727	0.0024	0.0951	0.0325	0.0172	0.0112
B3 s46	1000	0.576	0.0999	0.0427	0.0728	0.0027	0.1024	0.0441	0.0213	0.0154
B2 LF s46	915	0.580	0.1005	0.0440	0.0666	0.0046	0.1141	0.0649	0.0266	0.0212
B3 LF s46	915	0.549	0.0943	0.0331	0.0629	0.0228	0.1616	0.1460	0.0538	0.0538

Table S4: PMRAM ($n = 1410$): all models. Recall per class: glioma / meningioma / pituitary / normal.

Model	Accuracy [95% CI]	F1	AUC	ECE	Recall per class
B2 s42	92.06 [90.64, 93.48]	92.04	97.31	0.037	91.4 / 93.6 / 100.0 / 82.7
B2 ens.	94.68 [93.47, 95.82]	94.66	99.25	0.061	93.5 / 97.6 / 100.0 / 87.4
B3 s42	92.34 [90.92, 93.69]	92.21	97.38	0.044	92.4 / 96.3 / 100.0 / 80.1
B3 ens.	94.82 [93.62, 95.96]	94.81	99.12	0.065	93.2 / 98.5 / 100.0 / 87.4
B2 LF ens.	94.47 [93.26, 95.60]	94.45	99.27	0.059	93.0 / 93.6 / 100.0 / 90.9
B3 LF ens.	94.04 [92.77, 95.25]	94.01	99.06	0.063	93.8 / 92.0 / 100.0 / 89.7
B3 s42 zeroplane	92.41 [90.99, 93.76]	92.29	97.36	0.045	92.4 / 96.3 / 100.0 / 80.4
B3 ens. zeroplane	94.89 [93.69, 96.03]	94.87	99.10	0.065	93.5 / 98.2 / 100.0 / 87.7

Table S5: AJBDS-2023: all models and protocols (patient-cluster bootstrap 95% CIs).

Model	Protocol	Tumor Dice	All Dice	Empty FP	Missed lesion
B2 LF ens.	P0	29.50 [20.84, 38.85]	65.35	22.18 [19.04, 25.50]	54.41
B2 LF ens.	P1	27.68 [18.92, 37.00]	67.45	18.72 [15.88, 21.83]	59.47
B2 LF ens.	P2a	20.95 [13.96, 29.72]	70.37	12.43 [10.37, 14.93]	68.06
B2 LF ens.	P2b	20.76 [12.69, 30.15]	73.14	8.63 [6.70, 11.03]	72.63
B2 ens.	P0	36.54 [27.73, 46.10]	63.06	27.71 [23.94, 31.67]	41.33
B2 ens.	P1	34.88 [25.96, 44.31]	65.26	24.16 [20.64, 28.03]	45.91
B2 ens.	P2a	25.94 [17.57, 35.42]	71.12	13.16 [10.66, 15.92]	58.75
B2 ens.	P2b	29.18 [20.34, 38.91]	72.00	13.10 [10.42, 15.94]	59.07
B2 s42	P0	36.48 [28.16, 45.23]	59.56	32.40 [29.00, 35.84]	39.25
B2 s42	P1	35.91 [27.62, 44.64]	60.27	31.26 [28.08, 34.49]	41.17
B2 s42	P2a	35.44 [27.40, 44.01]	61.35	29.64 [26.25, 33.16]	40.69
B2 s42	P2b	31.14 [22.77, 39.93]	65.19	22.96 [19.93, 26.21]	51.85
B3 LF ens.	P0	31.42 [22.16, 41.08]	65.88	22.12 [19.42, 25.28]	50.80
B3 LF ens.	P1	28.20 [19.22, 37.64]	67.87	18.32 [15.66, 21.62]	57.87
B3 LF ens.	P2a	20.84 [13.73, 29.47]	72.25	9.86 [8.04, 12.09]	67.50
B3 LF ens.	P2b	22.08 [13.99, 30.98]	73.23	8.97 [7.12, 11.25]	70.71
B3 ens.	P0	38.00 [29.30, 47.60]	61.26	30.64 [26.90, 34.44]	39.25
B3 ens.	P1	34.81 [26.35, 43.99]	63.57	26.42 [23.02, 29.99]	46.71
B3 ens.	P2a	24.82 [17.11, 34.31]	71.51	12.23 [9.67, 15.40]	60.75
B3 ens.	P2b	26.82 [18.76, 36.56]	72.38	11.76 [9.36, 14.64]	62.52
B3 s42	P0	38.35 [28.57, 48.13]	57.15	36.31 [32.73, 40.03]	36.28
B3 s42	P1	38.07 [28.42, 47.57]	57.47	35.78 [32.24, 39.54]	37.08
B3 s42	P2a	37.53 [27.66, 47.39]	59.23	33.21 [29.75, 36.72]	37.72
B3 s42	P2b	36.37 [26.23, 46.06]	61.59	29.64 [26.35, 32.91]	43.34

Table S6: Subgroup analysis of tumor-slice Dice, 5-seed ensembles, B3 − B2 (Holm-adjusted Wilcoxon p) and seed-matched differences (seeds 42–46).

Setting	Subgroup	n	B2	B3	Δ [95% CI] (pp)	p_H	Seed-level Δ (pp)
official	axial	346	85.13	85.19	+0.06 [-0.45, +0.51]	1.00	+0.93, -0.40, +0.28, +0.05, -0.54
official	coronal	257	86.54	86.37	-0.17 [-0.56, +0.22]	1.00	+0.08, -0.52, -0.70, -0.46, -0.14
official	sagittal	257	88.33	87.75	-0.58 [-1.32, -0.01]	1.00	-0.02, -0.23, -0.51, -0.52, -0.75
official	glioma	254	76.44	75.88	-0.56 [-1.53, +0.30]	1.00	+0.92, -0.93, +0.07, -0.74, -0.52
official	meningioma	306	93.97	94.16	+0.18 [+0.03, +0.42]	0.00	+0.38, +0.21, -0.51, +0.27, -0.75
official	pituitary	300	87.42	87.13	-0.29 [-0.56, -0.03]	0.01	-0.04, -0.53, -0.25, -0.43, -0.18
dedup	axial	346	82.95	82.91	-0.03 [-0.59, +0.57]	0.36	-0.24, +0.42, -1.09, -0.07, -1.72
dedup	coronal	257	85.70	85.45	-0.25 [-0.74, +0.20]	0.64	+2.44, -0.15, -0.24, -0.60, -1.88
dedup	sagittal	257	87.79	87.76	-0.04 [-0.37, +0.32]	0.36	+0.98, -0.63, -0.37, -0.66, +0.34
dedup	glioma	254	75.27	75.37	+0.10 [-0.68, +0.90]	0.64	+2.46, -0.09, -0.56, +0.08, -0.97
dedup	meningioma	306	91.52	91.29	-0.23 [-0.62, +0.14]	0.00	-0.05, +0.28, -1.13, -1.27, -1.67
dedup	pituitary	300	87.22	87.08	-0.13 [-0.48, +0.26]	0.13	+0.62, -0.39, -0.16, +0.06, -0.77

Table S7: AJBDS-2023 fine-tuning, pre-specified protocol (uniform sampling, 600 steps). 5-fold patient-grouped CV; k = training patients (0 = zero-shot); patient-cluster bootstrap 95% CIs.

Model	k	Tumor Dice	All Dice	Empty FP	Missed	B3 − B2 tumor Dice (pp)
B2	0	36.5 [28.2, 45.2]	59.6	32.4 [29.0, 35.8]	39.2	—
B2	1	10.2 [2.9, 19.4]	71.9	6.7 [1.8, 12.8]	73.5	—
B2	3	15.8 [6.1, 29.1]	73.8	6.0 [3.8, 8.5]	72.5	—
B2	6	14.1 [2.0, 29.7]	73.1	6.4 [2.7, 10.4]	79.1	—
B2	all	4.5 [0.5, 10.3]	71.8	4.8 [2.1, 7.7]	91.2	—
B3	0	38.4 [28.6, 48.1]	57.1	36.3 [32.7, 40.0]	36.3	+1.9 [-1.7, +4.8]
B3	1	8.3 [1.8, 15.7]	73.0	4.5 [1.3, 8.7]	80.4	-1.9 [-6.4, +2.8]
B3	3	14.3 [3.5, 28.5]	72.5	7.2 [3.3, 11.3]	77.8	-1.5 [-6.5, +6.0]
B3	6	13.2 [2.1, 27.4]	73.7	5.2 [3.0, 7.4]	80.3	-1.0 [-5.3, +3.1]
B3	all	13.6 [2.2, 26.3]	76.3	1.9 [0.9, 3.1]	81.4	+9.1 [+0.3, +17.9]

Table S8: AJBDS-2023 fine-tuning, post-hoc amended protocol (lesion-balanced sampling, 2000 steps). 5-fold patient-grouped CV; k = training patients (0 = zero-shot); patient-cluster bootstrap 95% CIs.

Model	k	Tumor Dice	All Dice	Empty FP	Missed	B3 − B2 tumor Dice (pp)
B2	0	36.5 [28.2, 45.2]	59.6	32.4 [29.0, 35.8]	39.2	—
B2	1	6.7 [1.9, 13.0]	72.2	4.9 [1.6, 9.2]	78.3	—
B2	3	14.6 [6.5, 23.7]	71.3	8.9 [5.6, 12.2]	75.6	—
B2	6	23.6 [11.0, 37.0]	69.7	14.2 [9.4, 18.9]	62.4	—
B2	all	23.0 [9.3, 37.5]	70.6	12.8 [5.7, 21.0]	64.9	—
B3	0	38.4 [28.6, 48.1]	57.1	36.3 [32.7, 40.0]	36.3	+1.9 [-1.7, +4.8]
B3	1	5.6 [1.3, 11.0]	72.1	4.8 [1.7, 8.9]	81.5	-1.1 [-2.9, +1.1]
B3	3	17.3 [6.8, 31.5]	72.3	8.6 [5.8, 11.8]	68.1	+2.8 [-3.5, +9.4]
B3	6	24.8 [11.6, 39.6]	75.3	7.1 [4.0, 10.4]	65.1	+1.1 [-3.3, +5.8]
B3	all	26.2 [14.0, 37.5]	77.0	5.4 [2.0, 10.6]	62.9	+3.2 [-3.9, +10.3]

References

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[3] J. Amin, M. A. Anjum, N. Gul, M. Sharif, and S. Kadry, "Clinically acquired new challenging dataset for brain SOL segmentation: AJBDS-2023," *Data Brief*, vol. 52, p. 109915, 2023.